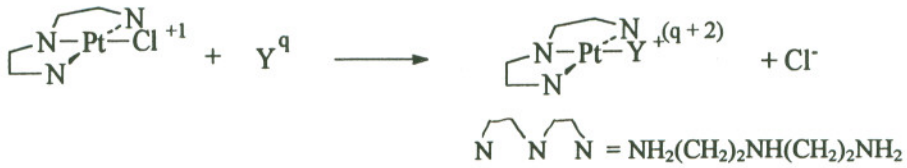
INORGANIC CUMULATIVE EXAM
December 5, 1998

1) The experimental rate law for the reaction below is:

Rate $= k_1[Pt(II)] + k_2[Pt(II)][Y^q]$

[ ] = molar conc. , q = charge on Y



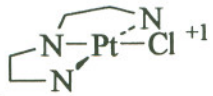
N N N = $NH_2(CH_2)_2NH(CH_2)_2NH_2$

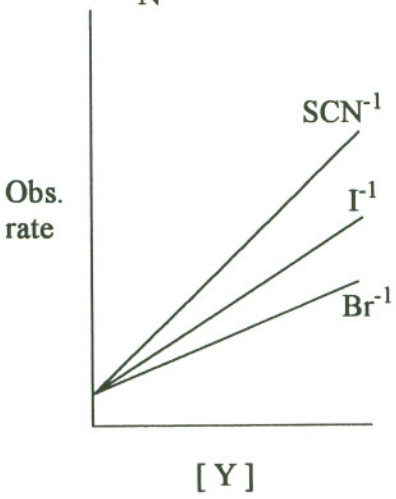
Account for the two terms in the rate law and describe a reasonable mechanism for this reaction.

2) What experimental tests might you apply to distinguish whether the mechanism for the reaction above involves associative or dissociative activation?

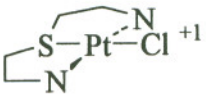
3) Account for the reactivity difference in the substitution reactions (displacement of Cl^{-1}) for the related complexes below.

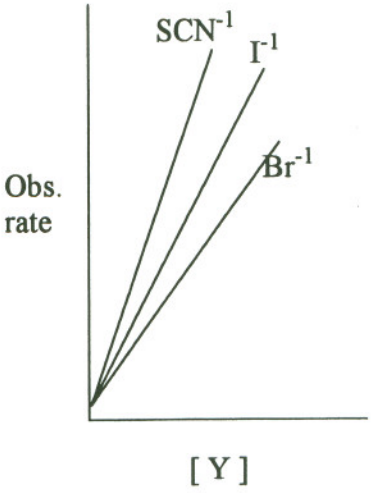
N N N = $NH_2(CH_2)_2NH(CH_2)_2NH_2$

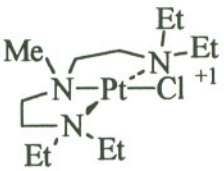




N S N = $NH_2(CH_2)_2S(CH_2)_2NH_2$





4) The rate of chloride substitution by pyridine is much slower for $[Pt\{Et_2N(CH_2)_2NMe(CH_2)2NEt_2\}Cl]^{+1}$ than for the complex in question 1. How does this substantiate or refute the mechanism you assigned in the question above.

$[Pt\{Et_2N(CH_2)_2NMe(CH_2)2NEt_2\}Cl]^{+1}$ =